A THREE-QUARTER EXCEPTIONAL-SET BOUND FOR THE ERDŐS–STRAUS EQUATION: AN INTERNALLY PROVED RESEARCH NOTE

ANONYMOUS

Abstract. **Archived version 1** (repository commit `ace654cf`). Superseded by the reorganised version C2 in `paper/es-threequarter-note.tex`; kept because other internal notes cite statements of this version by number.

This note, **internally proved but not externally refereed**, records the internally reviewed argument for

$$E(N) \ll N \exp\{-c(\log N)^{3/4}\},$$

where $E(N)$ counts denominators $n \le N$ for which $4/n$ is not a sum of three positive unit fractions. Power-sized multipliers supply cubic logarithmic class mass after an incidence-congestion pruning and Bombieri–Vinogradov. A fixed family of congruence classes is then assembled on the integers, using exact CRT probabilities, a small-prime selector, and an even Bonferroni polynomial. The complete coefficient and rounding ledger costs $\exp\{O((\log X)^4)\}$. All proof ingredients and their uniformities are given below; the three external-referee checkpoints are retained explicitly, and Section 5 gives a shorter route to the class supply (Cauchy–Schwarz against Bombieri–Vinogradov) that removes the distinct-prime-factor cutoff, the congestion pruning, and the unweighted incidence-moment lemma (Lemma 4.1) from the proof. The argument has undergone the internal checks described in Theorem 1.1; these do not constitute external refereeing, and "internally proved" means exactly that. This is not a proof of the Erdős–Straus conjecture, nor a claim of publication priority.

1. Statement, status, and parameter order

For a positive integer n, call n *representable* if there exist positive integers x_1, x_2, x_3 with

$$\frac{4}{n} = \frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3}. \tag{1}$$

The denominators x_i need not be distinct. Put

$$E(N) = \#\{1 \le n \le N : n \text{ is not representable}\},$$
$$E_{\mathrm{pr}}(N) = \#\{p \le N : p \text{ prime and not representable}\}.$$

The contribution of $n = 1$ is harmless. The conjecture asks for no exceptions for $n \ge 2$; a density bound does not establish that assertion.

Theorem 1.1 (internally proved; not externally refereed)**.** *There are absolute $c_*, C_* > 0$ such that, for every real $N \ge 3$,*

$$E_{\mathrm{pr}}(N) \le E(N) \le C_* N \exp\{-c_*(\log N)^{3/4}\}.$$

Date: INTERNALLY PROVED; internal checks only, not externally refereed.

The argument for this statement has undergone internal checks only. The internal record comprises a self-review ([N, §39.7]), a blind construction and reconciliation taking Theorem 34.8 there as input ([N, §41]; the blindness is self-attested, not independently audited), a standalone internal audit (the repository file `reviews/es-threequarter-note-review.md`*, which found no load-bearing gap and recorded repairs), a checkpoint re-derivation ([N, §76.4]), and a hostile review of Section 5 (the file* `reviews/wave32-sec76-review.md`*). These records are not three independent full-chain hostile reviews, and the theorem has not been verified by an external expert.*

This note follows [N, §§16,34,39]; [N, §41] is an internal blind re-derivation, not an independent published source. We use the reviewed weaker-bound note [LL] as follows: its Lemma 2.1 supplies the multiplier identity; Lemmas 3.1 and 3.2 supply elementary lattice sums and the multiplier harmonic sum; the distinctness argument is that of its Lemma 4.1; its Section 5 fixes our large-sieve conventions; and its Section 6 supplies the final semigroup argument. These ingredients are restated, and their proofs are reproduced or lightly adapted, so no consultation of a working notebook is needed to follow the calculation. We do *not* import [LL, Lemma 4.1]'s prime-counting conclusion outside its stated polylogarithmic multiplier range. That extension is the work of Section 4.

Here is the order of constants. Fix once and for all

$$\begin{gathered} 0<\kappa<1/240, \quad t=\log X, \quad K=\lfloor X^{\kappa}\rfloor, \quad H=K^{10}, \\ \mathcal{K}(K)=\{1\le k\le K : k\equiv 1 \pmod 4\}, \quad L_K=\operatorname{lcm}_{k\in\mathcal{K}(K)} k. \end{gathered} \tag{2}$$

An absolute sufficiently large K_0 is chosen in the lattice estimates. The distinct-prime-factor cutoff D is a fixed absolute integer chosen in Lemma 3.3. Constants in the analytic estimates are absolute once D is fixed, with thresholds allowed to depend on κ, D. In Section 7 we choose an absolute small $\eta>0$, obtain a positive void constant $c_v=c_v(\kappa,D)$, and then fix a sufficiently large $B=B(\kappa,D)$ for $y=Bt^3$. The Bonferroni constant $D_B=D_B(\kappa,D,B)$ is distinct from D. Finally choose $C_0=C_0(\kappa,D,B,D_B)$ for the ledger and $\alpha=\alpha(C_0)>0$ for $t=\alpha(\log N)^{1/4}$. All subsequent sufficiently-large thresholds have only these dependencies; no constant depends on a residue, subfamily, moment order, or N. Fixing, for example, $\kappa=1/480$ makes the final constants absolute. Constants are not asserted to be effective: standard Bombieri–Vinogradov is used without an effectivity refinement. This D-dependent list describes the retained original route. For the simplified route of Section 5, omit D and its choice, use the family $\mathcal{A}'_X$, and delete D from all dependencies; the remaining choices occur in the same order: κ, K_0, η and the void constant, B, D_B, C_0, and α.

The central distinction is between using prime distribution at scale X to *construct* many classes and counting those classes on *all integers* up to N. A prime-progression error $N/(\log N)^A$, with fixed A, is much larger than the desired $N\mathrm{e}^{-ct^3}$ in the final window. Such an error is never used at scale N. Exact integer counts, each with error at most one, replace it.

2. Forced classes and a fixed atom family

Lemma 2.1 (multiplier identity; [LL, Lemma 2.1])**.** *Let k,ℓ be positive integers with $k\ell\equiv 3 \pmod 4$, and let $A=(k\ell+1)/4=uvw$ be any factorization into positive*

integers. If $n \geq 1$ satisfies $nv \equiv -u \pmod{k\ell}$, then $s = (nv+u)/(k\ell)$ is a positive integer and

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsvw} + \frac{1}{nuvw}.$$

In particular every positive integer in the class $-uv^{-1} \pmod{k\ell}$ is representable; the inverse exists since $(A, k\ell) = 1$.

Proof. On the common denominator $nsuvw$ the three numerators sum to $nv+u+s = sk\ell + s = s(k\ell+1) = 4suvw$. Positivity and integrality of s follow directly from the hypotheses. □

For a precise endpoint convention take $x_j = 2^j X^{1/2}$ and use the disjoint blocks

$$I_j = (x_j, \min(2x_j, X)], \qquad 0 \leq j < \left\lceil \frac{\log X}{2\log 2} \right\rceil.$$

The last may be partial. In a block put $z_j = x_j^{1/6}$; we discard the last block for lower estimates if it is partial and enlarge it to a full block for upper estimates. There are $\asymp t$ full blocks and $\log x_j \asymp t$. Because $\kappa < 1/240$, for all sufficiently large X each block satisfies

$$z_j \geq X^{1/12} > K^{20} = H^2, \qquad \Lambda_j := \log(z_j/H) \geq \tfrac{1}{2}\log z_j. \tag{3}$$

Let $\mathcal{A}_X$ consist of all quadruples $A = (k, \ell, u, v)$ such that

$$\begin{gathered} k \in \mathcal{K}(K), \quad \ell \in I_j \text{ prime}, \quad H < u, v \leq z_j, \\ (u, v) = (uv, k) = 1, \quad \omega(uv) \leq D \log\log X, \quad 4uv \mid k\ell + 1. \end{gathered} \tag{4}$$

Here $\omega(m)$ counts distinct prime factors. Divisibility already forces $\ell \equiv 3 \pmod 4$. The atom denotes the cylinder

$$E_A = \{n : n \equiv -uv^{-1} \pmod{k\ell}\}, \qquad H_X(n) = \sum_{A \in \mathcal{A}_X} \mathbf{1}_{E_A}(n). \tag{5}$$

This is a fixed, unpruned, residue-independent family. Pruning will be used only to prove a lower mass bound within a fibre, never to define the polynomial on the integers.

Lemma 2.2 (deduplication and conditional independence)**.** *For every sufficiently large X in (2) the following hold. Every exceptional integer avoids every atom. At a fixed ℓ all atoms have distinct projections modulo ℓ. Consequently a compatible set of distinct atoms has distinct large primes ℓ_i, and its exact CRT probability is*

$$\mathbb{P}\left(\bigcap_i E_{A_i}\right) = \frac{\mathbf{1}_{\text{compatible}}}{\operatorname{lcm}(k_1, \ldots, k_j) \prod_i \ell_i}. \tag{6}$$

Conditional on any fixed residue $c \pmod{L_K}$, the coordinates $n \pmod \ell$ are independent and uniform. An atom is active in that fibre precisely when $u + cv \equiv 0 \pmod k$.

Proof. The identity proves avoidance and reduction modulo k proves the activation condition. If two atoms at the same $\ell \in I_j$ have the same projection, then $\ell \mid uv' - u'v$, but $|uv' - u'v| < z_j^2 < \ell$. Thus $uv' = u'v$, and both fractions being reduced gives $(u, v) = (u', v')$. Now $k\ell \equiv k'\ell \equiv -1 \pmod{4uv}$. Since $(\ell, 4uv) = 1$, we get $k \equiv k' \pmod{4uv}$; as $4uv > 4H^2 > K$, this means $k = k'$. In particular there are at most $z_j^2 \leq \ell^{1/3}$ atoms at that prime. This is the distinctness argument of [LL,

Lemma 4.1]. Finally all $\ell > X^{1/2} > K$ are distinct primes coprime to L_K. The Chinese remainder theorem proves both the density and independence assertions. Incompatible intersections are empty. □

All probability spaces below are finite and exact. When the selector S_y is present we use the uniform residue n modulo

$$\mathcal{M} = \operatorname{lcm}(L_K, P_y) \prod_{\substack{X^{1/2}<\ell\le X \\ \ell \text{ prime}}} \ell, \qquad P_y = \prod_{p\le y} p, \quad 2 \le y < X^{1/2}. \tag{7}$$

Without the selector one can take $P_y = 1$. A congruence whose modulus divides $\mathcal{M}$ has probability the reciprocal of its modulus. The possibly enormous size of $\mathcal{M}$ is a device for defining probabilities, not a partition of $[1, N]$ that will be counted term by term. The factor 24 used in the notebooks may be included in $\mathcal{M}$ without changing any argument; no multiplier here requires it.

3. Analytic inputs and the power-range lattice estimates

We record the external inputs precisely. For $q \ge 1$ and $(a, q) = 1$, Brun–Titchmarsh in the Montgomery–Vaughan form [MV] gives

$$\pi(V; q, a) \le \frac{2V}{\varphi(q)\log(V/q)} \qquad (V > q). \tag{8}$$

Only $q \le 4x^{1/3}$, $V = 2x$ is needed. For every fixed $R > 0$, Bombieri–Vinogradov [D, Chapter 28] supplies constants $A_R, C_R, x_R > 0$ such that, for all $x \ge x_R$,

$$\sum_{q\le x^{1/2}(\log x)^{-A_R}} E_x^*(q) \le C_R x(\log x)^{-R}, \tag{9}$$

where

$$E_x^*(q) = \max_{(a,q)=1} \left|\pi(2x; q, a) - \pi(x; q, a) - \frac{\operatorname{li}(2x) - \operatorname{li}(x)}{\varphi(q)}\right|.$$

The two endpoints follow from the usual maximal version with only a factor two in the constant. In particular $4x^{1/3}$ is below the level for every fixed R and all sufficiently large x.

We also use Shiu [S, Theorem 1, pp. 162–163]. If F is nonnegative and multiplicative, $F(p^a) \le A_1^a$ for every prime power, and for each $\epsilon > 0$ there is $A_2(\epsilon)$ with $F(n) \le A_2(\epsilon)n^\epsilon$, then for fixed $0 < \alpha_s, \beta_s < 1/2$,

$$\sum_{\substack{V-Y<n\le V \\ n\equiv a \pmod q}} F(n) \ll_{\alpha_s,\beta_s,F} \frac{Y}{\varphi(q)\log V} \exp\left\{\sum_{p\le V,\ p\nmid q} \frac{F(p)}{p}\right\} \tag{10}$$

uniformly for $q \ge 2$, $(a, q) = 1$, $q < Y^{1-\alpha_s}$ and $V^{\beta_s} < Y \le V$. The constant includes the growth data of F. In our applications $F(n) = a_0^{\omega(n)}$ for fixed $1 < a_0 < 2$, or $F(n) = n/\varphi(n)$. Both satisfy these hypotheses. For the first use $F(n) \le \tau(n)^{\log_2 a_0} \ll_{a_0,\epsilon} n^\epsilon$; for the second use the divisor expansion below. Fix boxes $(U, (1 + \delta_0)U]$ with $\delta_0 = 1/20$, and take $V = (1 + \delta_0)U$, $Y = \delta_0 U$ and $\alpha_s = \beta_s = 1/4$. For every $q \le K$ and $U \ge H = K^{10}$, both inequalities hold with room to spare once $K \ge K_0$. The unrestricted $q = 1$ upper estimates needed here follow elementarily from $F = 1 * h$: for these two functions h is nonnegative and supported on squarefree integers, with $h(p) = a_0 - 1$ and $1/(p - 1)$ respectively.

Indeed $\sum_{n\le V} F(n) \le V\sum_{d\le V} h(d)/d$; the Euler product gives $O_{a_0}(V(\log V)^{a_0-1})$ and $O(V)$. Thus no endpoint convention about modulus one in Shiu is required.

For $k \ge 1$, $(c,k)=1$, and $1 \le H < z$, put $\Lambda = \log(z/H)$ and set

$$\mathcal{S}(k,c;H,z) = \left\{(u,v): \begin{array}{l} H < u,v \le z,\ (u,v)=1,\ (v,k)=1, \\ u \equiv -cv \pmod{k} \end{array}\right\}.$$

Here $(u,k)=1$ follows automatically.

Lemma 3.1 (elementary lattice sums; [LL, Lemma 3.1])**.** *There exist absolute $K_0, C_1 > 0$ such that for every real $K \ge K_0$, every integer $1 \le k \le K$, every $(c,k)=1$, and every $z \ge H^2$ with $H = K^{10}$,*

$$\sum_{\mathcal{S}(k,c;H,z)} \frac{1}{uv} \ge \frac{1}{4}\frac{\varphi(k)}{k^2}\Lambda^2, \tag{11}$$

$$\sum_{\mathcal{S}(k,c;H,z)} \frac{1}{\varphi(u)\varphi(v)} \le C_1\frac{\varphi(k)}{k^2}\Lambda^2. \tag{12}$$

There is no polylogarithmic restriction on K.

Proof. We recall the elementary proof to make this uniformity explicit. For $H' \ge 1$, $z' > H'$, $\Lambda' = \log(z'/H')$, and $(a,k)=1$, integral comparison and Möbius inversion give

$$\sum_{\substack{H'<u\le z' \\ u\equiv a\ (k)}} \frac{1}{u} = \frac{\Lambda'}{k} + O(1/H'), \tag{13}$$

$$\sum_{\substack{H'<v\le z' \\ (v,k)=1}} \frac{1}{v} = \frac{\varphi(k)}{k}\Lambda' + O(\tau(k)/H'). \tag{14}$$

These include empty progressions. Write $T(H',z')$ for the double sum without $(u,v)=1$. For $\Lambda' \ge 1$,

$$T(H',z') = \frac{\varphi(k)}{k^2}\Lambda'^2 + O(\Lambda'\tau(k)/H').$$

Inverting $(u,v)=1$, using $(d,k)=1$ for a common divisor, and separating $d \le H$ from $d > H$, gives

$$\sum_{\mathcal{S}} \frac{1}{uv} = \frac{\varphi(k)}{k^2}\Lambda^2 \sum_{\substack{d\le H \\ (d,k)=1}} \frac{\mu(d)}{d^2} + O\left(\frac{\Lambda\tau(k)\log z}{H} + \frac{\Lambda^2}{H}\right).$$

For the tail use $T(H/d, z/d) \le (1+\log z)^2$ and $\log z \le 2\Lambda$. The main coefficient is $\prod_{p\nmid k}(1-p^{-2}) + O(H^{-1}) \ge 6/\pi^2 - O(H^{-1})$. Since $\tau(k)k^2/\varphi(k) \le k\tau(k)^2 \le 4K^2$, the relative error is $O(K^2/H) = O(K^{-8})$. This proves (11).

For the upper estimate drop $(u,v)=1$ and use

$$\frac{1}{\varphi(m)} = \frac{1}{m}\sum_{d\mid m}\frac{\mu^2(d)}{\varphi(d)}.$$

Applying (13) for $d \le H$ and the trivial harmonic bound for $d > H$ shows, uniformly for reduced a,

$$\sum_{\substack{H<u\le z\\ u\equiv a\ (k)}} \frac{1}{\varphi(u)} \ll \frac{\Lambda}{k} + \frac{\log H}{H} + \frac{\Lambda}{\sqrt{H}}.$$

Here $\sum_d \mu^2(d)/(d\varphi(d)) < \infty$, $\sum_{d\le H} \mu^2(d)/\varphi(d) \ll \log H$, and $\sum_{d>H} 1/(d\varphi(d)) \ll H^{-1/2}$. For clarity the middle estimate follows by another use of the divisor expansion and the convergent first sum; the tail follows from $\varphi(d) \gg \sqrt{d}$. These elementary bounds also prove convergence of the first sum. Similarly (14) gives

$$\sum_{\substack{H<v\le z\\ (v,k)=1}} \frac{1}{\varphi(v)} \ll \frac{\varphi(k)}{k}\Lambda + \frac{\tau(k)\log H}{H} + \frac{\Lambda}{\sqrt{H}}.$$

Since $\tau(k) \le 2\sqrt{K}$, $\varphi(k)/k \ge 1/(2\sqrt{K})$ and $\Lambda \ge 10\log K$, the secondary terms are absorbed uniformly. Multiply the last two estimates, applying the first with $a = -cv$, to obtain (12). □

Lemma 3.2 (multiplier harmonic sum; [LL, Lemma 3.2])**.** *For every real $K \ge 5$,*

$$h(\mathcal{K}(K)) := \sum_{k\in\mathcal{K}(K)} \frac{\varphi(k)}{k^2} = \frac{2}{\pi^2}\log K + O(1), \tag{15}$$

with absolute error constant. For every prime $p \le K$,

$$\sum_{\substack{k\le K\\ p\mid k}} \frac{\varphi(k)}{k^2} \le \frac{1+\log K}{p}. \tag{16}$$

Proof. Expand $\varphi(k)/k = \sum_{d\mid k} \mu(d)/d$. The progression $k \equiv 1 \pmod 4$ gives

$$h(\mathcal{K}(K)) = \sum_{\substack{d\le K\\ d \text{ odd}}} \frac{\mu(d)}{d^2}\left(\frac{1}{4}\log\frac{K}{d} + O(1)\right).$$

Now $\sum_{d \text{ odd}} \mu(d)/d^2 = 8/\pi^2$ and $\sum_d (1+\log d)/d^2 < \infty$. The second assertion follows by $\varphi(k)/k^2 \le 1/k$ and writing $k = pm$. □

Lemma 3.3 (low-ω mass in the full power range)**.** *There is a fixed absolute integer $D \ge 1$ such that, for all $K \ge K_0$, $H = K^{10}$, $z \ge H^2$ sufficiently large, $k \le K$ and $(c,k) = 1$,*

$$\sum_{\substack{(u,v)\in\mathcal{S}(k,c;H,z)\\ \omega(uv)\le D\log\log z}} \frac{1}{uv} \ge \frac{1}{8}\frac{\varphi(k)}{k^2}\Lambda^2. \tag{17}$$

The threshold and constants are absolute once D is fixed. Thus in (2), uniformly over every $\mathcal{J} \subseteq \mathcal{K}(K)$ and every $(c, L_{\mathcal{J}}) = 1$, summing (17) over $k \in \mathcal{J}$ gives a lower bound $\gg (\log z)^2 h(\mathcal{J})$, also with cutoff $D\log\log X$ when $z \le X$. Here $L_{\mathcal{J}} = \operatorname{lcm}_{k\in\mathcal{J}} k$ and $h(\mathcal{J}) = \sum_{k\in\mathcal{J}} \varphi(k)/k^2$.

Proof. Fix $a_0 = 3/2$. Apply (10) in the boxes described above to $F(n) = a_0^{\omega(n)}$. For each reduced v the u-congruence is reduced; summing the second variable over all

reduced classes gives, after division by UV, a bound

$$C\frac{1}{\varphi(k)}(\log U\log V)^{a_0-1}\exp\{-2a_0\sum_{p\mid k}p^{-1}\}.$$

The ratio of its local coefficient to $\varphi(k)/k^2$ has logarithm

$$\sum_{p\mid k}\{-2\log(1-1/p)-2a_0/p\}\le C_{a_0}\sum_{p}p^{-2};$$

the linear terms are nonpositive. All primes dividing k are below the box endpoints, so none is lost from this calculation. Drop $(u,v)=1$ for this upper estimate, cover the interval by full boxes (enlarging the last), and sum $O(\Lambda^2)$ box pairs. It follows that

$$\sum_{\mathcal{S}}\frac{a_0^{\omega(u)+\omega(v)}}{uv}\ll_{a_0}\frac{\varphi(k)}{k^2}\Lambda^2(\log z)^{2(a_0-1)}. \tag{18}$$

On the coprime pairs $\omega(uv)=\omega(u)+\omega(v)$. The omitted tail is consequently at most the right side of (18) multiplied by $(\log z)^{-D\log a_0}$. Fix any integer D with $D\log(3/2)>1$ (any larger fixed D also works) and enlarge the threshold until this tail is at most one eighth of $\varphi(k)\Lambda^2/k^2$. Lemma 3.1 then proves the claim. No application of Shiu has growing function parameters or an unreduced residue, and no step requires K to be a power of a logarithm. □

4. Congestion pruning and cubic prime-slice mass

For $(c,L_{\mathcal{J}})=1$, define

$$r_{\mathcal{J}}(u,v;c)=\#\{k\in\mathcal{J}:k\mid u+cv\}.$$

If $(u,v)=1$ and $k\mid u+cv$, reducedness of c modulo k implies $(uv,k)=1$. Thus this incidence counts precisely the multiplier choices in the lattice, before the prime progression is imposed.

Lemma 4.1 (second incidence moment; notes Lemma 34.7)**.** *There is an absolute constant C_2 such that for every $K\ge K_0$, $H=K^{10}$, $z\ge H^2$, every $\mathcal{J}\subseteq\mathcal{K}(K)$ and every $(c,L_{\mathcal{J}})=1$,*

$$\sum_{\substack{H<u,v\le z\\(u,v)=1}}\frac{r_{\mathcal{J}}(u,v;c)^2}{uv}\le C_2(\log z)^2(1+\log K)^3. \tag{19}$$

Proof. Expand the square. A pair k,k' requires the single congruence $d=[k,k']\mid u+cv$, with $(c,d)=1$ and $d\le K^2$. The same calculation (13)–(14), now at modulus d and without the coprimality restriction on the pair, gives

$$\sum_{\substack{H<u,v\le z\\(u,v)=1,\ d\mid u+cv}}\frac{1}{uv}\ll\frac{\varphi(d)}{d^2}(\log z)^2+\frac{\tau(d)(\log z)^2}{H}. \tag{20}$$

Indeed the original coprime pairs have $(v,d)=1$, so dropping $(u,v)=1$ while retaining $(v,d)=1$ is an upper bound. The direct harmonic calculation gives even $O(\Lambda\tau(d)/H)$ here; the weaker displayed error is the box-boundary estimate in notes (34.15). This does not misapply Lemma 3.1 with floor $(K^2)^{10}$.

For the main coefficients,

$$\sum_{k,k'\le K}\frac{\varphi([k,k'])}{[k,k']^2}\le\sum_{k,k'\le K}\frac{(k,k')}{kk'}\le\sum_{d\le K}\frac{1}{d}(1+\log(K/d))^2\ll(1+\log K)^3.$$

Here writing $k=da,k'=db$, $(a,b)=1$, and then dropping this last condition proves the middle bound. The total errors in (20) are $O(K^3(\log z)^2/H)$, because $\tau([k,k'])\le 2K$ and there are at most K^2 pairs. Since $H=K^{10}$ these errors are absorbed in (19). □

We state the unused hypothesis to distinguish precisely what pruning proves. The literal $H_{k\mathrm{BV}}(\kappa)$ of notes (34.2) asserts the existence of a function $\rho_\kappa(X)\to 0$ such that, for all sufficiently large X, simultaneously for $X^{1/2}\le x\le X$, $K_0\le K\le X^\kappa$, every $1\in\mathcal{J}\subseteq\mathcal{K}(K)$ and every reduced $c \pmod{24L_{\mathcal{J}}}$,

$$\sum_{k\in\mathcal{J}}\sum_{\substack{(u,v)\in\mathcal{S}(k,c;H,z)\\ \omega(uv)\le D\log\log X}}|E(x;4uv,-k^{-1})|\le\rho_\kappa(X)x\log x\,h(\mathcal{J}),\tag{21}$$

where $H=K^{10}$, $z=x^{1/6}$ and E is the error defining E_x^*. Every triple retains its multiplicity. The reducedness modulo $24L_{\mathcal{J}}$ can be replaced in the harvesting proof by reducedness modulo $L_{\mathcal{J}}$: since $L_{\mathcal{J}}$ is odd, every such class lifts to a reduced class modulo $24L_{\mathcal{J}}$, without changing any multiplier congruence. This hypothesis is *not* assumed or proved. Plain BV would multiply (9) by as much as $K(\log X)^{D\log 2}$, which cannot be absorbed for $K=X^\kappa$ by a fixed logarithmic saving.

Theorem 4.2 (pruned prime slice; notes Theorem 34.8)**.** *Fix $0<\kappa<1/240$ and D from Lemma 3.3. There are $a_h,A_h>0$ and $X_h=X_h(\kappa,D)$ such that for every $X\ge X_h$, every $K_0\le K\le X^\kappa$, every $1\in\mathcal{J}\subseteq\mathcal{K}(K)$, and every $(c,L_{\mathcal{J}})=1$, the following holds. Within each canonical block keep only triples with*

$$\begin{gathered}(u,v)\in\mathcal{S}(k,c;H,z_j),\quad k\in\mathcal{J},\quad\omega(uv)\le D\log\log X,\\ r_{\mathcal{J}}(u,v;c)\le T_X:=t^4,\quad 4uv\mid k\ell+1.\end{gathered}\tag{22}$$

Let $f^{\mathrm{good}}_{c;\mathcal{J}}(\ell)$ count their distinct classes modulo ℓ. Then uniformly in all these data,

$$a_ht^2h(\mathcal{J})\le\sum_{X^{1/2}<\ell\le X}\frac{f^{\mathrm{good}}_{c;\mathcal{J}}(\ell)}{\ell}\le A_ht^2h(\mathcal{J}).\tag{23}$$

Moreover (21) restricted by $r_{\mathcal{J}}\le T_X$ is $o(x\log x\,h(\mathcal{J}))$ uniformly, without assuming $H_{k\mathrm{BV}}$.

Proof. In a full block put $z=x^{1/6}$. By (3) and Lemma 3.3 the low-ω harmonic triple mass is $\gg(\log z)^2h(\mathcal{J})$. The mass removed by the congestion condition is at most

$$\frac{1}{t^4}\sum_{u,v}\frac{r_{\mathcal{J}}(u,v;c)^2}{uv}\ll\frac{(\log z)^2(1+\log K)^3}{t^4}=o((\log z)^2h(\mathcal{J})).\tag{24}$$

Uniformity for sparse subfamilies uses $h(\mathcal{J})\ge 1$ from $1\in\mathcal{J}$; indeed the relative error is $O_\kappa(1/t)$ even in that worst case. This proves the negligible loss of *harmonic* mass. To obtain the same assertion for the actual prime main-term weight, use the elementary Mertens consequence $n/\varphi(n)\ll\log\log(3n)$: split its prime product at $\log(3n)$; the smaller primes cost $O(\log\log(3n))$ and the larger ones a bounded factor. Since $4uv\le 4X^{1/3}$, the removed $1/\varphi(4uv)$ mass is at most $O(\log t)$ times

the removed harmonic mass. Relative to the original main-term mass, which is $\gg (\log z)^2 h(\mathcal{J})$, its loss is $O_\kappa(\log t/t) = o(1)$. Thus both weighted masses are controlled; no assertion is made that the discarded *actual prime count* is small.

Each retained triple asks that

$$\ell \equiv -k^{-1} \pmod{4uv}, \qquad q = 4uv \le 4x^{1/3}. \tag{25}$$

The residue is reduced and already is $3 \pmod 4$. For a fixed $q = 4uv$, coprimality allows at most $2^{\omega(uv)}$ ordered allocations of its prime powers to u, v. Each allocation has at most T_X admissible multipliers. Hence its entire retained multiplicity is

$$W_{\mathrm{good}}(q) \le t^{4+D\log 2}. \tag{26}$$

Choose once and for all $R > 4 + D\log 2 + 10$ in (9). Since $\log x \asymp t$ and $q \le 4x^{1/3}$ lies below the BV level, grouping by q bounds the whole retained triple error by $O_D(x(\log x)^{-10})$. All constants and thresholds here are independent of $c, \mathcal{J}, K, x$ in their specified ranges.

For the main term use $\mathrm{li}(2x) - \mathrm{li}(x) \ge x/\log(2x)$ and $1/\varphi(4uv) \ge 1/(4uv)$. Equation (24) therefore gives at least $ax \log x\, h(\mathcal{J})$ retained incidences. Lemma 2.2 proves that these are genuinely distinct classes for each ℓ; in particular repeated occurrences of a prime are not an overcount of a single class. Dividing by $2x$ gives $\gg \log x\, h(\mathcal{J})$ reciprocal mass in each full block.

For the upper bound drop both prunings. Brun–Titchmarsh and $\varphi(4uv) \ge 2\varphi(u)\varphi(v)$ for coprime u, v give

$$\sum_{x<\ell\le 2x} f_{c;\mathcal{J}}(\ell) \ll \frac{x}{\log x} \sum_{k\in\mathcal{J}} \sum_{\mathcal{S}(k,c;H,z)} \frac{1}{\varphi(u)\varphi(v)} \ll x \log x\, h(\mathcal{J})$$

by Lemma 3.1. Divide by x and sum over the blocks. There are $\asymp t$ full blocks for the lower bound; enlarging the last block gives the upper bound. This proves all assertions. □

Corollary 4.3 (uniform fibre masses)**.** *For every $X \ge X_h$ with $K = \lfloor X^\kappa \rfloor$ and every residue $c \pmod{L_K}$, put*

$$\mathcal{J}_c = \{k \in \mathcal{K}(K) : (k, c) = 1\}, \qquad \mu_c = \sum_\ell f_c(\ell)/\ell,$$
$$f_c(\ell) = \#\{A \in \mathcal{A}_X : \ell_A = \ell,\ k_A \mid u_A + cv_A\}.$$

There are constants $a_h, C_u > 0$, depending at most on κ, D, such that

$$a_h t^2 h(\mathcal{J}_c) \le \mu_c \le A_h t^2 h(\mathcal{J}_c) \le C_u t^3 \tag{27}$$

uniformly in all *c, not only reduced residues modulo L_K. For reduced c the mass is $\asymp_\kappa t^3$.*

Proof. An active atom necessarily has $(k, c) = 1$, since its residue modulo k is a unit. Thus active atoms have multipliers in $\mathcal{J}_c$, which contains 1, and $(c, L_{\mathcal{J}_c}) = 1$. The lower bound of Theorem 4.2 applies to this very subfamily, and its canonical good triples are contained in $\mathcal{A}_X$. The unpruned BT upper bound in that proof applies with the same $\mathcal{J}_c$. Finally use Lemma 3.2 and $\log K \le \kappa t$. For reduced c, $\mathcal{J}_c = \mathcal{K}(K)$ and $\log K \ge \kappa t/2$ once X is large. This checks explicitly the data-dependent-subfamily quantifiers. □

5. Unpruned supply by Cauchy–Schwarz (simplification)

This section, added after the independent re-derivation recorded in [N, §76], gives Theorem 4.2 for the *full* triple family, with no cutoff on $\omega(uv)$ and no congestion pruning. With it, Lemma 3.3, Lemma 4.1 as a pruning device, the constant D, and the cutoff T_X are no longer used anywhere in the proof of Theorem 1.1 (Corollary 5.3). Section 4 is retained as the reviewed original route.

Lemma 5.1 (weighted second incidence moment)**.** *There is an absolute constant C_3 such that for every $K \ge K_0$, $H = K^{10}$, $z \ge H^2$, every $\mathcal{J} \subseteq \mathcal{K}(K)$ and every $(c, L_{\mathcal{J}}) = 1$,*

$$\sum_{\substack{H<u,v\le z\\(u,v)=1}} \frac{2^{\omega(uv)}\, r_{\mathcal{J}}(u,v;c)^2}{\varphi(4uv)} \le C_3 (\log z)^4 (1+\log K)^3. \tag{28}$$

Proof. For coprime u, v one has $\varphi(4uv) \ge 2\varphi(u)\varphi(v)$: if uv is odd this is an equality, and if $u = 2^a m$ with $a \ge 1$, m odd, then $\varphi(4u) = 2^{a+1}\varphi(m) = 4\varphi(u)$. Expand the square as in Lemma 4.1: a pair $k, k' \in \mathcal{J}$ imposes the single congruence $d = [k,k'] \mid u + cv$ with $d \le K^2$ and $(c,d) = 1$. For coprime u, v with $d \mid u + cv$ one has $(v,d) = 1$, so dropping $(u,v) = 1$ while keeping $(v,d) = 1$ is an upper bound, and the residue $u \equiv -cv \pmod d$ is reduced. Thus each pair contributes at most

$$\tfrac{1}{2} \sum_{\substack{H<v\le z\\(v,d)=1}} \frac{2^{\omega(v)}}{\varphi(v)} \sum_{\substack{H<u\le z\\ u\equiv -cv \ (d)}} \frac{2^{\omega(u)}}{\varphi(u)}.$$

Cover $(H, z]$ by consecutive boxes $(U, (1+\delta_0)U]$ starting at $U = H$, enlarging the last box when necessary; all endpoints are then at most $(21/20)z$, which affects no logarithmic bound. For $d \ge 2$ apply (10) to $F(n) = 2^{\omega(n)} n/\varphi(n)$, which is multiplicative with $F(p^a) = 2p/(p-1) \le 4 \le 4^a$ and $F(n) \le 4^{\omega(n)} \le \tau(n)^2 \ll_\epsilon n^\epsilon$, in these full boxes, with modulus $d \le K^2 \le U^{1/5}$ and a reduced residue. For $d = 1$ (the pair $k = k' = 1$, always present since $1 \in \mathcal{J}$), the nonnegative convolution $F = 1 * h$ below gives $\sum_{n\le V} F(n) \ll V \log V$, hence each box contributes at most $U^{-1} \sum_{n\le (21/20)U} F(n) \ll \log U$, which is the same estimate since $\varphi(1)/1^2 = 1$. Since $\sum_{p\le V} F(p)/p = 2\log\log V + O(1)$ and $\exp\{-\sum_{p\mid d} 2/(p-1)\} \asymp (\varphi(d)/d)^2$ (the logarithm of the ratio is $\sum_{p\mid d} O(p^{-2})$), the inner box sum of $2^{\omega(u)}/\varphi(u) \le F(u)/U$ is $\ll (\log U)\varphi(d)/d^2$. There are $O(\log z)$ boxes, so the inner sum is $\ll (\log z)^2 \varphi(d)/d^2$, uniformly in the residue. For the outer sum drop $(v,d) = 1$; $F = 1 * h$ with $h \ge 0$ supported on squarefree integers, $h(p) = (p+1)/(p-1)$ (indeed $h(p^a) = F(p^a) - F(p^{a-1}) = 0$ for $a \ge 2$), gives $\sum_{n\le V} F(n) \ll V \log V$ and, by partial summation, $\sum_{H<v\le z} 2^{\omega(v)}/\varphi(v) \ll (\log z)^2$. Hence each pair contributes $\ll (\log z)^4 \varphi(d)/d^2$, and $\sum_{k,k'\le K} \varphi([k,k'])/[k,k']^2 \ll (1+\log K)^3$ exactly as in Lemma 4.1. □

For $1 \in \mathcal{J} \subseteq \mathcal{K}(K)$ and $(c, L_{\mathcal{J}}) = 1$ let $f^{\mathrm{all}}_{c;\mathcal{J}}(\ell)$ count the triples (k,u,v) with $k \in \mathcal{J}$, $(u,v) \in \mathcal{S}(k,c;H,z_j)$ for the block $I_j \ni \ell$, and $4uv \mid k\ell + 1$. By Lemma 2.2, whose distinctness argument uses coprimality, $u, v \le z_j$ with $z_j^2 < \ell$, $H < u, v$ with $4H^2 > K$, and $4uv \mid k\ell + 1$, but no ω cutoff, these triples have pairwise distinct residues modulo ℓ, so f^{all} also counts distinct classes.

Theorem 5.2 (unpruned prime slice)**.** *Fix $0 < \kappa < 1/240$. There are $a_h, A_h > 0$ and $X_h = X_h(\kappa)$ such that for every $X \ge X_h$, every $K_0 \le K \le X^{\kappa}$, every $1 \in \mathcal{J} \subseteq \mathcal{K}(K)$ and every $(c, L_{\mathcal{J}}) = 1$,*

$$a_h t^2 h(\mathcal{J}) \le \sum_{X^{1/2} < \ell \le X} \frac{f^{\mathrm{all}}_{c;\mathcal{J}}(\ell)}{\ell} \le A_h t^2 h(\mathcal{J}). \tag{29}$$

Proof. The upper bound is the last paragraph of the proof of Theorem 4.2, which used no pruning. For the lower bound fix a full block $(x, 2x]$ with $z = x^{1/6}$, so that $4uv \le 4x^{1/3}$ and $\Lambda \ge \frac{1}{2}\log z \asymp t$ by (3). Then

$$\sum_{x < \ell \le 2x} f^{\mathrm{all}}_{c;\mathcal{J}}(\ell) = \sum_{k \in \mathcal{J}} \sum_{(u,v) \in \mathcal{S}(k,c;H,z)} \big[\pi(2x; 4uv, -k^{-1}) - \pi(x; 4uv, -k^{-1})\big],$$

the residue being reduced since $(k, 4uv) = 1$. Write each bracket as $(\mathrm{li}(2x) - \mathrm{li}(x))/\varphi(4uv) + E(x; 4uv, -k^{-1})$. By (11) the main term is at least

$$\frac{x}{\log(2x)} \sum_{k \in \mathcal{J}} \sum_{\mathcal{S}(k,c;H,z)} \frac{1}{4uv} \ge \frac{x}{16 \log(2x)} \Lambda^2 h(\mathcal{J}) \gg x\, t\, h(\mathcal{J}).$$

For the error, group the triples by $q = 4uv$ and put $W_c(q) = \sum_{(u,v):\, 4uv = q} r_{\mathcal{J}}(u, v; c)$, the sum over coprime $H < u, v \le z$ with $uv = q/4$; there are at most $2^{\omega(q/4)}$ such ordered pairs, so by Cauchy–Schwarz $W_c(q)^2 \le 2^{\omega(q/4)} \sum_{4uv=q} r_{\mathcal{J}}(u, v; c)^2$. A second Cauchy–Schwarz gives

$$\Big|\sum_{\text{triples}} E\Big| \le \sum_q W_c(q) E^*_x(q) \le \Big(\sum_q W_c(q)^2 E^*_x(q)\Big)^{1/2} \Big(\sum_{q \le 4x^{1/3}} E^*_x(q)\Big)^{1/2}.$$

For $q \le 4x^{1/3}$, (8) and $\mathrm{li}(2x) - \mathrm{li}(x) \le x/\log x$ give $E^*_x(q) \ll x/(\varphi(q)\log x)$, so by Lemma 5.1

$$\begin{aligned} \sum_q W_c(q)^2 E^*_x(q) &\ll \frac{x}{\log x} \sum_{\substack{H < u, v \le z \\ (u,v) = 1}} \frac{2^{\omega(uv)} r_{\mathcal{J}}(u, v; c)^2}{\varphi(4uv)} \\ &\ll \frac{x}{\log x} (\log z)^4 (1 + \log K)^3 \ll x\, t^6, \end{aligned}$$

using $\log z \asymp \log x \asymp t$ and $\log K \le \kappa t$. With the fixed choice $R = 26$ in (9), $\sum_{q \le 4x^{1/3}} E^*_x(q) \le C_{26} x (\log x)^{-26}$ for $x \ge x_{26}$. The total error is therefore $\ll x\, t^{-10}$, which is $o(x\, t\, h(\mathcal{J}))$ because $h(\mathcal{J}) \ge 1$ from $1 \in \mathcal{J}$. Divide by $2x$ and sum over the $\asymp t$ full blocks. All constants depend only on κ and on the absolute Shiu, Brun–Titchmarsh and Bombieri–Vinogradov constants; none depends on c, $\mathcal{J}$, K or x. □

Corollary 5.3 (the theorem without D, pruning, or Lemma 4.1)**.** *Let $\mathcal{A}'_X$ be the atom family (4) with the condition $\omega(uv) \le D \log\log X$ deleted. Then Lemma 2.2, Corollary 4.3, Theorem 6.3 (conditional-independence proof), Lemma 7.1, Theorem 8.2 and Section 9 hold with $\mathcal{A}_X$ replaced by $\mathcal{A}'_X$, Theorem 4.2 replaced by Theorem 5.2, the references to a pruned lower estimate replaced by the unpruned supply estimate, and D deleted from all parameter dependencies; and Theorem 1.1 follows with the same order for the surviving parameters and without any use of Lemma 3.3, Lemma 4.1, the lower W_k calculation of Lemma 6.1, the ordered-atom replay, D, or T_X.*

Proof. In the original route, the cutoff and the congestion threshold are used to establish low-ω mass (Lemma 3.3), to control the retained main-term mass (24), and to bound the Bombieri–Vinogradov multiplicity (26); the optional lower W_k calculation in Lemma 6.1 also uses the cutoff. Theorem 5.2 replaces the entire pruned-supply argument. The proof of the main theorem then uses only the conditional-independence proof of Theorem 6.3, not Lemma 6.1 or the ordered-atom replay. Lemma 2.2 does not use the cutoff. Corollary 4.3 uses the lower bound of the supply theorem for the subfamily $\mathcal{J}_c \ni 1$ with $(c, L_{\mathcal{J}_c}) = 1$, and the unpruned upper bound; both are (29). The conditional-independence proof of Theorem 6.3 needs only Corollary 4.3; Lemma 7.1 needs only the fibre lower bound and the selector computation; Theorem 8.2 needs the moment bound, the void bound, and the crude inventory (48), which never used the cutoff. The transfer to all denominators is untouched. □

Remark 5.4. The device is the one used for the fixed-multiplier theorem in [LL, Section 4]: a multiplicity as large as X^{κ} at one modulus q is harmless when the error is controlled in mean square, because it is paid for by its own weight $1/\varphi(q)$ in $\sum_q W_c(q)^2/\varphi(q)$. In the power range the weight $2^{\omega(uv)}$ makes the allocation count explicit, and Shiu's theorem for $2^{\omega(n)}n/\varphi(n)$ costs two extra powers of $\log z$, which the fixed logarithmic saving of Bombieri–Vinogradov absorbs. The unused hypothesis (21) is now irrelevant to the proof rather than merely unassumed. The exponent, the cubic mass per fibre, the Bonferroni degree and the ledger are unchanged.

6. Residue profiles and factorial moments

This section retains the residue-resolved estimate and the local induction of notes §§39.2–39.3, including the referee-sensitive factors. We also give the conditional-independence formulation of the same moment bound; Corollary 4.3 makes its uniform upper hypothesis explicit.

For $k \in \mathcal{K}(K)$, $g \mid k$, and a reduced residue $a \pmod g$, define

$$W_{k,a}(g) = \sum_{\substack{A\in\mathcal{A}_X,\ k_A=k\\ -u_Av_A^{-1}\equiv a\ (g)}} \frac{1}{k\ell_A}, \qquad W_k = \sum_{\substack{A\in\mathcal{A}_X\\ k_A=k}} \frac{1}{k\ell_A}.$$

The sole class modulo $g = 1$ is understood as reduced.

Lemma 6.1 (residue-resolved upper bound; notes (39.11))**.** *For fixed κ, D there are $a_p, A_p > 0$ and $X_p = X_p(\kappa, D)$ such that for every $X \ge X_p$, every $k \in \mathcal{K}(K)$, every $g \mid k$ and every reduced $a \pmod g$,*

$$W_{k,a}(g) \le \frac{A_p t^2}{\varphi(g)} \frac{\varphi(k)^2}{k^3}, \qquad a_p t^2 \frac{\varphi(k)^2}{k^3} \le W_k \le A_p t^2 \frac{\varphi(k)^2}{k^3}. \tag{30}$$

In particular $\mu_X := \sum_A 1/(k_A\ell_A) \asymp_\kappa t^3$.

Proof. Fix a block and put $q_0 = \operatorname{lcm}(g, \operatorname{rad} k) \le k$. In boxes $u \asymp U$, $v \asymp V$, the conditions $(uv, k) = 1$ and $-uv^{-1} \equiv a \pmod g$ permit exactly $\varphi(q_0)^2/\varphi(g)$ ordered pairs of unit classes modulo q_0. Indeed reduction of units modulo q_0 onto units modulo g is surjective, with all fibres of size $\varphi(q_0)/\varphi(g)$. Apply Shiu to $F(n) = n/\varphi(n)$ in each class. Its prime value is $p/(p-1)$, whence Mertens' estimate

gives

$$\frac{1}{\log U}\exp\left\{\sum_{p\le(1+\delta_0)U,\ p\nmid q_0}\frac{1}{p-1}\right\}\ll\exp\left\{-\sum_{p\mid k}\frac{1}{p-1}\right\}\asymp\frac{\varphi(k)}{k}.$$

The last comparability is uniform: the logarithm of its ratio is a sum of $O(p^{-2})$. Shiu applies since $q_0\le K\le U^{1/10},V^{1/10}$, with the interval conditions already checked in Section 3. Drop only $(u,v)=1$ for an upper bound. Multiplying the two one-class estimates by the number of compatible pairs and dividing by UV gives

$$\sum_{\text{box}}\frac{1}{\varphi(u)\varphi(v)}\ll\frac{1}{\varphi(g)}\left(\frac{\varphi(k)}{k}\right)^2.$$

Summing the $O((\log z)^2)$ box pairs proves the full residue estimate

$$\sum_{\substack{H<u,v\le z,\ (uv,k)=1\\ -uv^{-1}\equiv a\ (g)}}\frac{1}{\varphi(u)\varphi(v)}\ll\frac{(\log z)^2}{\varphi(g)}\frac{\varphi(k)^2}{k^2}. \tag{31}$$

No divisor factor or k^ϵ is suppressed. When applying BT we restrict back to coprime pairs, so that $1/\varphi(4uv)\le 1/(2\varphi(u)\varphi(v))$ is valid. Equation (8) implies

$$\sum_{\substack{x<\ell\le 2x\\ \ell\equiv -k^{-1}\ (4uv)}}\frac{1}{\ell}\ll\frac{1}{\varphi(4uv)\log x}.$$

Combine this with (31), divide by k and sum $O(t)$ blocks to get the first inequality of (30).

For completeness the lower W_k bound does not assume a polynomially small BV main term can survive a fixed logarithmic error. For a fixed k, sum Lemma 3.3 over the $\varphi(k)$ residues $c \pmod k$. Each reduced coprime pair occurs in exactly one of them, giving low-ω harmonic mass $\gg(\log z)^2\varphi(k)^2/k^2$ without a ratio condition. For this fixed k, a modulus has at most $t^{D\log 2}$ ordered allocations; there is no factor K. BV with fixed $R>D\log 2+10$ makes its error $O_D(x(\log x)^{-10})$. The elementary Mertens product implies $\varphi(k)/k\ge\prod_{p\le K}(1-1/p)\gg 1/\log K$ for $K\ge 3$. Thus the block main term $\gg x\log x\,(\varphi(k)/k)^2$ is uniformly $\gg_\kappa x/t$, and dominates that error. Division by $2xk$ and summation over full blocks prove the lower bound.

Finally, writing $b(k)=(\varphi(k)/k)^2$, Cauchy–Schwarz and Lemma 3.2 give

$$\sum_{k\in\mathcal{K}(K)}\frac{b(k)}{k}\ge\frac{h(\mathcal{K}(K))^2}{\sum_{k\in\mathcal{K}(K)}1/k}\gg\log K;$$

the reverse inequality follows from $b(k)\le 1$. This proves the mass assertion. Notice that μ_X is an *unconditional cylinder mass*, whereas μ_c in (27) is a conditional prime-coordinate mass. They are not interchangeable term by term. □

At an odd prime p the nonconstant part of the proxy Euler factor for $b(k)/k$ is

$$\sum_{e\ge 1}\frac{(1-1/p)^2}{p^e}=\frac{p-1}{p^2}.$$

Its share of the full local factor is $(p-1)/(p^2+p-1)$. This is an exact *Euler-factor* identity, not an exact finite-K share of W_k, for which only comparability has been proved.

Define, for $y \ge 2$,

$$q_y(k) = \prod_{p \mid k,\ p \le y} \frac{p}{p-1}, \qquad b_y(g) = \prod_{p^e \| g,\ p \le y} \varphi(p^e) \prod_{p^e \| g,\ p > y} p^e.$$

Lemma 6.2 (uniform multiplier bound; notes Lemma 39.3)**.** *For every real $y \ge 2$, integer $R \ge 1$, and real $K \ge 3$,*

$$\sum_{k \in \mathcal{K}(K)} \frac{\varphi(k)^2}{k^3} q_y(k) \prod_{p \mid (k,R),\ p > y} \frac{p}{p-1} \ll \log K, \tag{32}$$

with an absolute constant independent of R, y.

Proof. Enlarge to all odd k and to their Euler product with $p \le K$. An unmodified nonconstant local mass is $(p-1)/p^2$; either permitted modification changes it to $1/p$. The ratio of the modified full factor to the original is $1 + O(p^{-2})$, with a convergent product. The original product is bounded by $\prod_{p \le K} (1 - 1/p)^{-1} \ll \log K$. All terms are nonnegative, so the product enlargement is legitimate. □

Theorem 6.3 (all upper factorial moments; notes Theorem 39.4)**.** *For fixed κ, D there are $C_F, X_F > 0$ such that, for every $X \ge X_F$, every real $2 \le y < X^{1/2}$, and* every *integer $m \ge 1$,*

$$\mathbb{E}((H_X)_m \mid S_y = 1) \le (C_F t^3)^m. \tag{33}$$

Here $S_y = \mathbf{1}_{(n, P_y)=1}$ and $(h)_m = h(h-1)\cdots(h-m+1)$. In particular the base C_F is independent of the order m.

Conditional-independence proof. Reveal $c \pmod{L_K}$, retaining the condition $S_y = 1$. Since $y < X^{1/2}$, this does not condition any large-prime coordinate. By Lemma 2.2, in this fibre H_X is a sum of independent Bernoulli variables I_ℓ with means $f_c(\ell)/\ell$; at most one atom at each ℓ can be hit. Expanding the falling factorial gives

$$\mathbb{E}((H_X)_m \mid c, S_y = 1) = \sum_{\ell_1, \dots, \ell_m \text{ distinct}} \prod_{i=1}^{m} \frac{f_c(\ell_i)}{\ell_i} \le \mu_c^m \le (C_u t^3)^m.$$

Corollary 4.3 applies to every c admitted by the selector. Averaging over these fibres proves (33) with $C_F = C_u$. This conditioning is solely in the CRT probability space and creates no finite-interval coefficient cost. □

For transparency we also replay the notes' ordered-atom proof, so the third referee checkpoint is not bypassed by the formulation just given. Given previous compatible atoms with multiplier lcm R, a new atom with multiplier k, $g = (k, R)$, and a new ℓ, has relative probability

$$\frac{q_y(k)}{k\ell} b_y(g) \tag{34}$$

if its unit residue agrees modulo g, and zero otherwise. Here is the prime-power check. If $p^e \| k$ and $p^f \| R$ (allow $f = 0$), the new local probability, subject to agreement, is

$f \ge e$	1
$0 < f < e$	$p^{-(e-f)}$
$f = 0,\ p \le y$	$1/\varphi(p^e)$
$f = 0,\ p > y$	p^{-e}.

For $p \le y$ the factor in (34) is $[p/(p-1)]p^{-e}\varphi(p^{\min(e,f)})$ when $f > 0$; this is precisely $p^{\min(e,f)-e}$. The other cases follow immediately. Unequal prime-power exponents are thus included, not replaced by radicals.

Compatibility fixes one unit residue modulo g. Apply Lemma 6.1 to its atom sum, multiply by $q_y(k)b_y(g)$, and use the exact cancellation

$$\frac{b_y(g)}{\varphi(g)} = \prod_{p\mid g,\ p>y} \frac{p}{p-1}. \tag{35}$$

The sum of the relative probabilities of all permissible new atoms is at most

$$A_p t^2 \sum_{k\in\mathcal{K}(K)} \frac{\varphi(k)^2}{k^3} q_y(k) \prod_{p\mid(k,R),\ p>y} \frac{p}{p-1} \ll t^2 \log K \ll_\kappa t^3$$

by Lemma 6.2. Excluding old ℓ only decreases this sum. Distinct compatible atoms have distinct primes ℓ_i by Lemma 2.2, so each step contributes a fresh factor $1/\ell$. Iteration over ordered compatible distinct atoms proves (33) again, with another fixed base constant. Discarding consistency would instead charge shared primes by their full lcm-collapse factors. That estimate is too large; a naive Janson argument is neither used nor justified here.

7. An exponentially small conditioned void

The all-integer CRT distribution contains nonreduced multiplier fibres, even though exceptional primes larger than K are reduced. One cannot replace this distribution by a uniform distribution on reduced residues without paying for that replacement. We use the exact small-prime selector

$$y = Bt^3, \qquad S_y(n) = \mathbf{1}_{(n,P_y)=1} = \sum_{d\mid P_y} \mu(d)\mathbf{1}_{d\mid n}. \tag{36}$$

For any fixed B, eventually $2 \le y < K < X^{1/2}$. Its expansion has $2^{\pi(y)}$ terms, with $\pi(y) \ll y/\log y$ and $\log P_y = \vartheta(y) \ll y$ by Chebyshev's elementary bounds. Every exceptional prime larger than y satisfies $S_y = 1$.

Lemma 7.1 (conditioned c-free void; notes Lemma 39.5)**.** *For fixed κ, D there are $B_0 > 0$ and $c_v > 0$ such that, for every fixed $B \ge B_0$ and all $X \ge X_v(\kappa, D, B)$,*

$$\mathbb{P}(H_X = 0 \mid S_y = 1) \le \mathrm{e}^{-c_v t^3}, \qquad y = Bt^3. \tag{37}$$

One may choose c_v depending only on κ, D, uniformly in fixed $B \ge B_0$ after allowing the threshold to depend on B.

Proof. Reveal $c \pmod{L_K}$ and retain the selector condition. Put

$$Z(c) = \sum_{\substack{y<p\le K,\\ p\mid c}} \sum_{p\mid L_K} \frac{1}{p}. \tag{38}$$

The restriction $p \mid L_K$ is important: otherwise divisibility by p is not defined on the residue c. In particular not every prime $p \equiv 3 \pmod 4$ up to K divides an admissible multiplier. For supported $p > y$ the events $p \mid c$ are independent with probabilities

$1/p$, also after conditioning on $S_y = 1$. Since $0 < y/p < 1$ and $\mathrm{e}^u - 1 \le (\mathrm{e}-1)u$ on $[0,1]$,

$$\mathbb{E}(\mathrm{e}^{yZ} \mid S_y = 1) = \prod_{y<p\le K,\ p\mid L_K} \left(1 + \frac{\mathrm{e}^{y/p}-1}{p}\right)$$
$$\le \exp\left\{(\mathrm{e}-1)y \sum_{p>y} p^{-2}\right\} \le \mathrm{e}^{C_Z}, \tag{39}$$
$$\mathbb{P}(Z > \eta \mid S_y = 1) \le \mathrm{e}^{-\eta y + C_Z}. \tag{40}$$

Here C_Z is absolute; even summing over all integers $> y$ bounds the exponent by an absolute constant.

Choose absolute $a_0 > 0$, K_1 such that $h(\mathcal{K}(K)) \ge a_0 \log K$ for $K \ge K_1$ by Lemma 3.2. The selector rules out every supported $p \le y$ dividing c. By (16) and a union bound over the remaining prime divisors,

$$h(\mathcal{J}_c) \ge h(\mathcal{K}(K)) - \sum_{y<p\le K,\ p\mid L_K,\ p\mid c} \sum_{k\le K,\ p\mid k} \frac{\varphi(k)}{k^2} \ge (a_0 - 2Z(c)) \log K$$

for K large. Fix $0 < \eta \le a_0/4$. On $Z(c) \le \eta$ this gives $h(\mathcal{J}_c) \ge (a_0/2)\log K \ge a_0 \kappa t/4$. Theorem 4.2 applies to this arbitrary subfamily because $1 \in \mathcal{J}_c$ and $(c, L_{\mathcal{J}_c}) = 1$; no reducedness modulo all of L_K is being assumed.

Conditional independence and distinctness now give the exact product for the unpruned active family, followed by its pruned lower estimate:

$$\mathbb{P}(H_X = 0 \mid c, S_y = 1) = \prod_{\ell} (1 - f_c(\ell)/\ell)$$
$$\le \exp\{-\mu_c\} \le \exp\{-a_h t^2 h(\mathcal{J}_c)\} \le \mathrm{e}^{-a_h t^3 a_0 \kappa/4} \tag{41}$$

on the good event. The equality remains valid if the selector has prime coordinates not in L_K, as all of them are below $X^{1/2}$ and independent of the ℓ-coordinates. Put $a_v = a_h a_0 \kappa/4$ and choose B_0 with $\eta B_0 \ge 2a_v$. Averaging (41) and using (40) gives $\mathbb{P}(H_X = 0 \mid S_y = 1) \le \mathrm{e}^{-a_v t^3} + \mathrm{e}^{-2a_v t^3 + C_Z}$. For sufficiently large X, this is at most $\mathrm{e}^{-a_v t^3/2}$. Take $c_v = a_v/2$. □

The role of the selector is quantitative. A merely Markov bound for the lost multiplier mass would be far too weak. The exponential parameter $y \asymp t^3$ in (39) makes the bad fibres as rare as the required saving, while its own congruence expansion still has acceptable size. Revealing c above is not an expansion into the L_K possible integer progressions; that distinction is crucial in the next section.

8. Bonferroni assembly on the integers

Lemma 8.1 (an even Bonferroni majorant)**.** *For every even integer $r \ge 0$ and every integer $h \ge 0$, let $Q_r(h) = \sum_{j=0}^{r} (-1)^j \binom{h}{j}$, with $\binom{h}{j} = 0$ for $j > h$. Then*

$$Q_r(0) = 1, \quad Q_r(h) = \binom{h-1}{r} \ge 0 \quad (h \ge 1),$$
$$\mathbf{1}_{h=0} \le Q_r(h) \le \mathbf{1}_{h=0} + \binom{h}{r+1}. \tag{42}$$

Proof. Pascal's identity telescopes the alternating sum to $(-1)^r\binom{h-1}{r}$ for $h \geq 1$. If $h \leq r$ this is zero; if $h \geq r+1$, then $\binom{h}{r+1} = h\binom{h-1}{r}/(r+1) \geq \binom{h-1}{r}$. The case $h = 0$ is immediate. □

Theorem 8.2 (critical-window assembly; notes Theorem 39.6)**.** *Fix κ, D and $B \geq B_0$ as above. There are positive constants c_a, D_B, C_0 and X_a, depending only on these fixed choices, such that for every $X \geq X_a$, if r is the least even integer at least $D_B t^3$, then*

$$\nu_X(n) = S_y(n)Q_r(H_X(n)) \geq 0, \qquad \mathbb{E}_{\mathrm{CRT}}\nu_X \leq \mathrm{e}^{-c_a t^3}. \tag{43}$$

It is at least one on every exceptional prime $n > \max(K, y)$. Its expansion into plain congruence classes has degree at most $r + \pi(y)$, maximum modulus $q_{\max}$ and total absolute coefficient sum T_{abs} satisfying

$$\begin{aligned} r + \pi(y) &= O_{B,D_B}(t^3), \\ q_{\max} &\leq P_y(KX)^r, \\ \log q_{\max} &\leq C_\vartheta B t^3 + r(1+\kappa)t, \\ T_{\mathrm{abs}} &\leq 2^{\pi(y)} \sum_{j=0}^{r} \binom{|\mathcal{A}_X|}{j}, \qquad \log T_{\mathrm{abs}} \leq C_L t^4. \end{aligned} \tag{44}$$

Here C_ϑ is an absolute Chebyshev constant and $C_L = C_L(\kappa, B, D_B)$. For every real N with $\log N \geq C_0 t^4$,

$$\sum_{1 \leq n \leq N} \nu_X(n) \ll_{\kappa,D,B,D_B} N\mathrm{e}^{-c_a t^3}. \tag{45}$$

Proof. By Lemma 8.1, Lemma 7.1, and Theorem 6.3,

$$\begin{aligned} \mathbb{E}(Q_r(H_X) \mid S_y = 1) &\leq \mathrm{e}^{-c_v t^3} + \frac{\mathbb{E}((H_X)_{r+1} \mid S_y = 1)}{(r+1)!} \\ &\leq \mathrm{e}^{-c_v t^3} + \left(\frac{\mathrm{e}C_F t^3}{r+1}\right)^{r+1}. \end{aligned} \tag{46}$$

Here $m! \geq (m/\mathrm{e})^m$ for all integers $m \geq 1$. Choose $D_B > \mathrm{e}C_F$ so large that $D_B \log(D_B/(\mathrm{e}C_F)) \geq 2c_v$. As $r + 1 \geq D_B t^3$, the last term is at most $\mathrm{e}^{-2c_v t^3}$. Increasing the threshold and setting $c_a = c_v/2$ gives (43), after multiplication by $\mathbb{P}(S_y = 1) \leq 1$. Positivity, and its value at an exceptional prime larger than y, follow from the identity lemma and (42).

We now account for *every* term, including the selector. The identity

$$\binom{H_X(n)}{j} = \sum_{\substack{\mathcal{B} \subseteq \mathcal{A}_X \\ |\mathcal{B}| = j}} \prod_{A \in \mathcal{B}} \mathbf{1}_{E_A}(n)$$

holds for all integers n. Expanding this and (36) produces terms with coefficients $(-1)^j\mu(d)$ for $d \mid P_y$. Each intersection is empty or one plain congruence class, by CRT, with modulus

$$q = \mathrm{lcm}(d, k_1\ell_1, \ldots, k_j\ell_j) \leq P_y(KX)^j. \tag{47}$$

In fact any selector divisor sharing a prime with an atom multiplier makes the term empty, because atom residues are units; ignoring that cancellation only enlarges our upper ledger. Every nonempty term modulus divides $\mathcal{M}$ in (7).

For a completely crude inventory, choose $k \le K$, $\ell \le X$ and $u, v \le X^{1/6}$. There are at most

$$|\mathcal{A}_X| \le KX^{4/3} \le \exp\{(4/3+\kappa)t\} \tag{48}$$

possibilities. Conditions of primality, coprimality and divisibility only reduce this bound. Therefore

$$\begin{aligned}\log T_{\mathrm{abs}} &\le \pi(y)\log 2 + \log(r+1) + r\log\max(1,|\mathcal{A}_X|)\\ &\le C_\pi B t^3 \log 2 + \log(r+1) + r(4/3+\kappa)t \le C_L t^4.\end{aligned} \tag{49}$$

An absolute C_π with $\pi(y) \le C_\pi y$ suffices in this line; the sharper $\pi(y) \ll y/\log y$ gives the stated selector degree. The exponent is $O(rt)$, not $O(r)$; we retain this loss explicitly. Equation (47) and Chebyshev give the modulus ledger. Since $r \le D_B t^3 + 2$, all bounds in (44) follow with fixed constants of the dependencies stated.

For every residue class $a \pmod q$, without any primality restriction,

$$\#\{1 \le n \le N : n \equiv a \pmod q\} = N/q + O(1),$$

with absolute error at most one. Summing with the signed coefficients, and bounding rounding errors by their absolute sum, yields

$$\sum_{n\le N} \nu_X(n) = N\mathbb{E}_{\mathrm{CRT}}\nu_X + O(T_{\mathrm{abs}}) \le N\mathrm{e}^{-c_a t^3} + \mathrm{e}^{C_L t^4}. \tag{50}$$

Choose C_0 greater than twice C_L and twice a fixed upper constant for $\log q_{\max}/t^4$. Then $\log N \ge C_0 t^4$ ensures both $T_{\mathrm{abs}} \le N^{1/2}$ and $q_{\max} \le N^{1/2}$. The latter is convenient but not necessary for the exact $O(1)$ count. For large X, $N^{1/2} \le N\mathrm{e}^{-c_a t^3}$, since $\frac{1}{2}\log N \ge (C_0/2)t^4 \ge c_a t^3$. This proves (45). □

This theorem does not establish the literal hypothesis $H'_{\mathrm{PF}}(k\ell;\mathrm{good})$ of notes §34.4. That hypothesis demands coefficient sum $\mathrm{e}^{O(r)}$ and a majorant of the larger, c-dependently pruned avoider predicate. Here the family is fixed and unpruned, its avoider is smaller but still contains every exception, and the ledger is honestly $\mathrm{e}^{O(rt)}$. The entire L_K-fibre partition is absent from (49). No assertion about a prime-counting mean for ν_X is required.

9. Exceptional primes and all denominators

Proof of Theorem 1.1 for primes. For large N, put $L = \log N$, take $t = \alpha L^{1/4}$ and $X = \mathrm{e}^t$, with fixed $0 < \alpha \le C_0^{-1/4}$. Every exceptional prime larger than $\max(K, y)$ is counted with weight at least one by ν_X, so

$$E_{\mathrm{pr}}(N) \le K + y + \sum_{n\le N} \nu_X(n) \ll K + y + N\exp\{-c_a\alpha^3 L^{3/4}\}.$$

The omitted range is negligible: $K \le \mathrm{e}^{\kappa\alpha L^{1/4}}$ and $y = B\alpha^3 L^{3/4}$, whereas the logarithm of the last main bound is $L - c_a\alpha^3 L^{3/4}$. Reducing the saving constant if needed gives $E_{\mathrm{pr}}(N) \ll N\mathrm{e}^{-c_p(\log N)^{3/4}}$ for a fixed $c_p > 0$. Increasing the outer constant covers bounded $N \ge 3$. □

Transfer to all denominators; [LL, Section 6]. If $p \mid n$ and $4/p = \sum_i 1/x_i$, then $4/n = \sum_i 1/((n/p)x_i)$. Every prime factor of an exceptional integer is therefore exceptional. Thus exceptional integers lie in the multiplicative semigroup generated

by exceptional primes, including the empty product 1. This containment, not its converse, is all that is used.

Put $g(u) = u^{3/4}$. Choose $c_0 > 0$ and $x_0 \geq \mathrm{e}$ such that for every $x \geq x_0$ the prime bound reads $E_{\mathrm{pr}}(x) \leq x\mathrm{e}^{-c_0 g(\log x)}$; a decrease of the saving constant absorbs its prefactor. Fix $0 < \eta_0 < c_0$, let $L = \log x$, and put $\delta = \eta_0 g(L)/L = \eta_0 L^{-1/4}$, which is less than $1/4$ for large x. Rankin's inequality gives

$$E(x) \leq x^{1-\delta} \prod_{p \leq x,\ p \text{ exceptional}} (1 - p^{-1+\delta})^{-1}. \tag{51}$$

The higher prime-power terms in the logarithm of the product are $O(\sum_p p^{-3/2}) = O(1)$ uniformly for $\delta < 1/4$. The finitely many primes below x_0 also cost a fixed constant. Partial summation gives

$$\sum_{x_0 < p \leq x,\ p \text{ exceptional}} p^{-1+\delta} \ll 1 + \mathrm{e}^{\delta L - c_0 g(L)} + \int_{\log x_0}^{L} \mathrm{e}^{\delta u - c_0 g(u)}\, du.$$

Since $g(u)/u = u^{-1/4}$ decreases, $\delta u \leq \eta_0 g(u)$ for $u \leq L$. The endpoint is bounded by one and the integral by $\int_{\log x_0}^{\infty} \mathrm{e}^{-(c_0-\eta_0)u^{3/4}}\, du < \infty$. The Euler product in (51) is bounded uniformly in x. Consequently $E(x) \ll x^{1-\delta} = x\mathrm{e}^{-\eta_0(\log x)^{3/4}}$. After fixing κ absolutely, all constants are absolute, completing the claimed argument. □

10. The heuristic mass-driven three-quarter ceiling

We record the large-sieve convention of [LL, Section 5], even though our interval transfer is Bonferroni rather than an invocation of that sieve. Montgomery [M, §8, p. 561] states, for an integer set in an interval of length Y avoiding $0 \leq \omega(\ell) < \ell$ classes at each prime in a finite set $\mathcal{P}$,

$$|\mathcal{N}| \leq \frac{Y + Q^2}{S(Q)}, \qquad S(Q) = \sum_{s \leq Q,\ s \mid \prod \mathcal{P}} \mu^2(s) \prod_{\ell \mid s} \frac{\omega(\ell)}{\ell - \omega(\ell)}.$$

This is the large sieve, not the distinct larger sieve. Writing $g(\ell) = \omega(\ell)/(\ell - \omega(\ell))$ and $G = \prod(1 + g(\ell))$, the Rankin tail with $v = 1/\log X$ gives, when $\ell \leq X$ and $\omega(\ell) \leq \ell/2$,

$$\frac{G - S(Q)}{G} \leq \exp\left\{ -\frac{\log Q}{\log X} + 2(\mathrm{e} - 1) \sum_{\ell} \frac{\omega(\ell)}{\ell} \right\}.$$

Thus mass $\mu \asymp t^{\mathcal{B}}$ and this product-tail budget balance $t\mu \lesssim \log N$. Our Bonferroni ledger has the same balance: $r \asymp \mu$ and $\log T_{\mathrm{abs}} = O(rt)$. The resulting exponent is $\theta = \mathcal{B}/(\mathcal{B}+1)$. Vaughan's prime-only supply has $\mathcal{B} = 2$; here $\mathcal{B} = 3$. (The symbol $\mathcal{B}$ in this paragraph is not the selector constant B.)

For completeness, the cubic total identity-class supply asserted in notes §18 has a short elementary verification. For $M \equiv 3 \pmod 4$, put $A = (M+1)/4$ and let $F(M)$ count the distinct multiplier-identity classes modulo M, using every factorization $A = uvw$ once as a source and then deduplicating residues. They are exactly $-4D \pmod M$ with $D \mid A^2$: one direction uses $D = u^2 w$; conversely at $p^e \| A$, $p^d \| D$, set the exponents in u, w, v to $\lfloor d/2 \rfloor$, $d - 2\lfloor d/2 \rfloor$, $e - \lfloor d/2 \rfloor - (d - 2\lfloor d/2 \rfloor)$. The divisors $D < A$ give distinct residues since $0 < 4D < M$, so

$$\tfrac{1}{2}(\tau(A^2) - 1) \leq F(M) \leq \tau(A^2).$$

The coefficient identity $\tau(n^2) = \sum_{d^2 \mid n} \mu(d)\tau_3(n/d^2)$ follows from $\sum_{e\ge 0}(2e+1)z^e = (1-z^2)/(1-z)^3$. Also $\sum_{m\le x}\tau_3(m) = \frac{1}{2}x\log^2 x + O(x\log x)$: sum $\lfloor x/(ab)\rfloor$ over $ab \le x$, whose total rounding error is $O(x\log x)$, and use the elementary harmonic sums for $\sum_{ab\le x} 1/(ab) = \frac{1}{2}\log^2 x + O(\log x)$. Summing over d and then partially summing yields

$$\sum_{n\le x}\frac{\tau(n^2)}{n} = \frac{\log^3 x}{6\zeta(2)} + O(\log^2 x), \qquad \sum_{M\le Q,\ M\equiv 3 \pmod 4}\frac{F(M)}{M} \asymp \log^3 Q.$$

Changing $1/(4A-1)$ to $1/(4A)$ costs only $O(1)$. Thus the entire identity-class supply, with distinct moduli and residues, already has exponent three; counting the same M again through another factorization $M = k\ell$ does not create new supply.

Remark 10.1 (the ceiling is a theorem for this architecture). The calculation above is heuristic, but its conclusion $\theta = \mathcal{B}/(\mathcal{B}+1) = 3/4$ is now a theorem for the architecture of this note [KA2, Cor. 6.1, Remark 5.4], [KA3, Cor. 4.2]; see also [SL, Theorem 10.14, Remark 10.15]. The majorant $\nu_X = S_y\, Q_r(H_X)$ of (43) is ≥ 0 on $\mathbb{Z}$ and ≥ 1 on the whole set of integers avoiding the atom classes and the classes $0 \bmod p$, $p \le y$; the atoms $-uv^{-1} \bmod k\ell$ are classes of $\mathcal{R}(k\ell)$ with $k\ell \le \ell^{1+2\kappa}$, where $\mathcal{R}(M) = \{-4D \bmod M : D \mid ((M+1)/4)^2\}$ (as above); the primes are $\le X \le N$; and the bound is $N\,\mathbb{E}\nu_X$ plus at most $T_{\mathrm{abs}} \le N^{1/2}$. For every method of exactly this form — a nonnegative CRT majorant that is ≥ 1 on the whole avoider set of a finite family of $\mathcal{R}(M)$-, (a,D)-, Case-A and selector classes with primes $\le N^{O(1)}$, used through $N\,\mathbb{E}\nu + \sum_i |a_i|$ with $\sum_i |a_i| < N$ — the saving is $\le C(\log N)^{3/4}$, with no condition on the moduli [KA3, Cor. 4.2], [SL, Theorem 10.14]. (For this note's atoms $k\ell \le \ell^{1+2\kappa}$, so $G \le P(G)^{1+\beta}$ with $\beta = 2\kappa < 1/120$, and the earlier bounded-β form [KA2, Thm 5.2] already applied; the factor $(\log\log N)^{3/4}$ of the earlier general form is removed in [KA3].) So the exponent $3/4$ of Theorem 1.1 is sharp for its method, up to constants: no gain $(\log N)^{3/4}\omega(N)$ with $\omega \to \infty$ is possible in this class. The same cap holds for every CRT-admissible large sieve with frequencies of polynomial level [SL, Theorem 14.2]. Not covered: large-sieve frequencies of super-polynomial level, direct counts and hybrid accounting above modulus $N/2$, majorants ≥ 1 only on $[1,N]$, non-CRT input (such as tuple counts with non-CRT main terms), other class types, primes beyond $N^{O(1)}$, and the other items of [SL, §16]. The Case-A part uses Elsholtz–Tao Prop. 1.4 and §7 (published, not re-proved); the part relevant here, $\mathcal{R}(M)$- and selector classes, uses only Shiu's theorem. These results are internally reviewed only. Nothing about the ceiling is used in the proof of Theorem 1.1.

11. The three external-referee checkpoints

The following are exactly the three mathematical checkpoints identified in notes §39.7, with their locations in this standalone presentation. Internal re-derivation is not external refereeing. Each checkpoint has now been re-derived independently, from the statements rather than by transcription ([N, §76.4]); the findings are recorded after each item. The theorem carries the label **internally proved; internal checks only, not externally refereed**, with the internal record described in Theorem 1.1.

(1) *Residue-resolved local factors.* Re-derive (31) (notes (39.11)) uniformly for every $k \le X^{\kappa}$, every divisor $g \mid k$, and every unit residue. The $q_0 = \mathrm{lcm}(g, \mathrm{rad}\, k)$ class count, Shiu interval hypotheses, and the factor

$\exp\{-2\sum_{p\mid k}(p-1)^{-1}\}$ must all be retained. The upper bound uses BT, not a BV main term below its error. Lemma 6.1 gives the full calculation and separately checks that its fixed-k lower main term does dominate the BV error. *Re-derivation:* confirmed. The checkpoint is not load-bearing: it enters only the ordered-atom replay of Theorem 6.3, whose conditional-independence proof needs only Corollary 4.3.

(2) *Pruned supply for the data-dependent subfamily.* Check Theorem 4.2 (notes Theorem 34.8) and its use with $\mathcal{J}_c$ in Corollary 4.3 and Lemma 7.1 (notes Lemma 39.5). This includes the power-range lattice and Shiu estimates, every second-moment boundary error, $1 \in \mathcal{J}_c$ for uniform pruning, the polylogarithmic modulus multiplicity after pruning, and the fact that the lower mass occurs in the *canonical* $z = x^{1/6}$ boxes. These boxes, not a larger per-prime cutoff, define our fixed atom family. The exponential bad-fibre calculation uses only primes supported by L_K. *Re-derivation:* confirmed line by line. The pruned-supply component is replaced by Theorem 5.2, i.e. by Lemma 5.1 (one Shiu application in progressions modulo $d \le K^2$ in boxes above K^{10}, plus the elementary modulus-one case), Cauchy–Schwarz, and Bombieri–Vinogradov with a fixed R; the elementary lattice estimates, the $\mathcal{J}_c$ quantifiers and fibre bounds, and the selector's exponentially small bad-fibre estimate remain required.

(3) *Prime-power consistency versus lcm collapse.* Replay (34)–(35) and the ordered moment induction (notes (39.14)–(39.19)), including unequal prime-power exponents. A shared conditioned p^e contributes $\varphi(p^e)$, an unconditioned one p^e; the residue share cancels $\varphi(p^e)$, leaving only $p/(p-1)$ in the latter case. Any missed factor would be raised through $r \asymp t^3$. The independent-Bernoulli proof of Theorem 6.3 provides a further check, using the uniform all-fibre upper bound rather than a Poisson approximation. *Re-derivation:* confirmed, including the exact conditional independence on $\mathbb{Z}/\mathcal{M}$, the distinctness of atoms at a common ℓ, the exact falling-factorial identity, the local prime-power table, and the $e^{O(t^4)}$ ledger.

Two further items outside the list were checked in the re-derivation: the Rankin transfer of Section 9 and the parameter inequalities (3) from $\kappa < 1/240$. Both are correct.

No step in the displayed implication is supplied by the unproved literal $H_{k\mathrm{BV}}$ or by a partition-free sieve hypothesis, and none invokes Elliott–Halberstam. The proof calculations above have been checked in preparing this condensation; no unresolved step has been silently replaced by an assumption. Nevertheless an external expert's reading of these checks and a primary-literature priority audit are still required before treating the theorem as externally established. The stronger universal sieve ceiling and an exact finite-truncation interpretation of the proxy Euler share are not asserted.

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